

**SLR-TX – 60**

Seat No.	
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T.E. (Civil Engineering) (Part – I) (Old-CGPA) Examination, 2018
TRANSPORTATION ENGINEERING – I

Day and Date : Monday, 10-12-2018

Max. Marks : 70

Time : 2.30 p.m. to 5.30 p.m.

- Instructions :** 1) Q. No. 1 is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. 3. **Each** question carries **one** mark.
2) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**
3) Figure on **right** indicates **full** marks.
4) Assume suitable data **wherever** necessary and mention it **clearly**.

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

14

- 1) Which type of bridge foundation is suitable when rock level close to the ground or top soil strata is strong ?

- | | |
|---------------------|------------------------|
| a) Deep foundations | b) Shallow foundations |
| c) Pile foundation | d) Well foundation |

- 2) Consider the following statements,

Excessive camber is not provided on the roads because

1. transverse tilt causes discomfort
2. of formation of cross ruts
3. of likely toppling over of highly laden bullock carts
4. of higher cost involved

Which of these statements are correct ?

- | | | | |
|---------------|---------------|---------------|---------------|
| a) 2, 3 and 4 | b) 1, 3 and 4 | c) 1, 2 and 4 | d) 1, 2 and 3 |
|---------------|---------------|---------------|---------------|

- 3) Ruling gradient on highways as per IRC in plain terrain is

- | | | | |
|------------|------------|-------------|-------------|
| a) 1 in 30 | b) 1 in 60 | c) 1 in 100 | d) 1 in 200 |
|------------|------------|-------------|-------------|

- 4) If the difference in elevation between the edges of a pavement of width 9 m and its crown is 7.5 cm, what is the camber of the pavement ?

- | | | | |
|------------|------------|------------|------------|
| a) 1 in 60 | b) 1 in 45 | c) 1 in 30 | d) 1 in 15 |
|------------|------------|------------|------------|

- 5) Bitumen grade 80/100 indicates that under the standard test conditions, penetration value of would vary from

- | | |
|-------------------|----------------------|
| a) 0.8 mm to 1 mm | b) 8 mm to 10 mm |
| c) 8 cm to 10 cm | d) 0.08 mm to 0.1 mm |

- 6) In cement concrete pavements, tie bars are installed in

- | | |
|---------------------|------------------------|
| a) Expansion joints | b) Contraction joints |
| c) Warping joints | d) Longitudinal joints |

P.T.O.



7) Match **List – I** (Item) with **List – II** (Use) and select the correct answer using the codes :

List – I

- A) O and D survey
- B) Collision diagram
- C) OMC
- D) Radius of relative stiffness

List – II

- 1) Concrete pavement design
- 2) Compaction
- 3) Accident survey
- 4) Traffic survey

Codes :

	A	B	C	D
a)	3	4	1	2
b)	4	3	2	1
c)	3	4	2	1
d)	4	3	1	2

8) Match **List – I** (Studies) with **List – II** (Purpose) and select the correct answer using the codes :

List – I

- A) Economic studies
- B) Financial studies
- C) Traffic studies
- D) Engineering studies

List – II

- 1) For road location and alignment
- 2) For population and agricultural pattern
- 3) For ascertaining the source of income
- 4) For traffic volume and traffic flow patterns

Codes :

	A	B	C	D
a)	1	2	3	4
b)	2	3	4	1
c)	3	4	2	1
d)	1	3	2	4

- 9) The corrected modulus of subgrade reaction for standard diameter plate is 6kg/cm^3 . What would be the modulus of subgrade reaction of the soil when tested with a 30 cm diameter plate ?
 a) 15kg/cm^3 b) 25kg/cm^3 c) 30kg/cm^3 d) 60kg/cm^3
- 10) Speed regulations on roads is decided on the basis of
 a) 60th percentile cumulative frequency
 b) 75th percentile cumulative frequency
 c) 80th percentile cumulative frequency
 d) 85th percentile cumulative frequency
- 11) For the construction of Water Bound Macadam (WBM) roads, the correct sequence of operations after spreading coarse aggregates is
 a) Dry rolling, wet rolling, application of screening and application of filler
 b) Dry rolling, application of filler, wet rolling and application of screening
 c) Dry rolling, application of screening, wet rolling, and application of filler
 d) Dry rolling, application of screening, application of filler and wet rolling,
- 12) Which one of these methods of tunnel construction is not suitable in rocks ?
 a) Full face method b) Compressed air method
 c) Heading and benching method d) Drift method
- 13) Which one of the following methods is generally considered the best for tunnel ventilation ?
 a) Driving a drift through the tunnel
 b) 'Blowin' method
 c) 'Blowout' method
 d) Combination of 'Blowin' and 'Blowout' methods
- 14) Which type of bridge is normally the longest type' ?
 a) Beam b) Arch c) Cable stayed d) Suspension

Set P



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T.E. (Civil Engineering) (Part – I) (Old-CGPA) Examination, 2018
TRANSPORTATION ENGINEERING – I

Day and Date : Monday, 10-12-2018
Time : 2.30 p.m. to 5.30 p.m.

Marks : 56

Instructions : 1) Figure on **right** indicates **full** marks.
2) Assume suitable data wherever necessary and mention it **clearly**.

SECTION – I

2. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain the role of transportation in the development of country and write the different modes of transport.
- b) The area of a certain district is 80000 sq. km and number of towns 86. Calculate the length of different categories of roads to be provided in this district by the year 2001. Overall road density aimed at 82 km per 100 sq.km area.
- c) Define Stopping Sight Distance (SSD) and calculate the stopping sight distance required on sloping gradient of – 2.5% for a design speed of 100 kmph. Assume a reaction time of 2.5 sec, coefficient of friction of 0.7 and brake efficiency of the vehicle is 50%. Also calculate the intermediate sight distance.
- d) Write a short notes on objectives of
 - 1) Super elevation and curve widening
 - 2) Transition curves.

3. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) What are the objectives of O and D study ? Enlist the methods of conducting O and D study.
- b) The average normal flow of traffic on cross roads A and B during design period are 400 and 250 PCU per hour, the saturation flow values on these roads are estimated as 1250 and 1000 PCU per hour respectively. The all-red time required for pedestrian crossing is 12 sec. Design the two phase traffic signal with pedestrian crossing by Webster's method. Sketch the phase diagram.
- c) What are the various tests carried out on bitumen ? Briefly mention the principle and use of each.
- d) Write a short notes on :
 - 1) Types of parking facilities
 - 2) Steps in bituminous mix design.

SECTION – II

4. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain 'Flexible Pavement and Rigid Pavement' and bring out the points of difference.
- b) Enumerate the construction steps of Water Bound Macadam (WBM) road.
- c) Calculate the stresses at the interior, edge and corner region of cement concrete pavement using Westergaard's analysis. Use the following data : Wheel Load $P = 4100$ kg, Modulus of elasticity of cement concrete $E = 0.3$ million kg/cm² pavement thickness, $h = 15$ cm, Poisson's ratio, $\mu = 0.15$, modulus of subgrade reaction, $K = 3.0$ kg/cm³, radius of contact area, $a = 15$ cm.

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- d) Write design steps of CBR method of flexible pavement design as per IRC guidelines. Calculate the CSA for the given data.
- 1) Type of road-two lane undivided carriageway
 - 2) Design CBR value – 5.0%
 - 3) Initial traffic on completion of construction = 300 cv per day
 - 4) Average growth rate = 6.0% per year
 - 5) Design life = 10 years
 - 6) VDF value = 2.5
 - 7) Lane distribution factor = 0.75.
5. Answer **any two** questions (7 marks **each**) : **(2×7=14)**
- a) What is scour and scour depth ? A bridge is proposed to be constructed across an alluvial stream carrying a discharge of 300 m³/sec. Assuming the value of silt factor = 1.1, determine the maximum scour depth when the bridge consist of three spans of 30 m each. Assume maximum scour depth = 2 d at noses of piers.
 - b) Describe drainage in tunneling.
 - c) State methods of tunneling in soft rock. Explain with sketch any one method.
 - d) Write short notes on :
 - 1) IRC class AA loading
 - 2) Bridge bearings.
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Seat No.	
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Set	Q
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T.E. (Civil Engineering) (Part – I) (Old-CGPA) Examination, 2018
TRANSPORTATION ENGINEERING – I

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Max. Marks : 70

Time : 2.30 p.m. to 5.30 p.m.

- Instructions :** 1) Q. No. 1 is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. 3. **Each** question carries **one** mark.
- 2) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**
- 3) Figure on **right** indicates **full** marks.
- 4) Assume suitable data **wherever** necessary and mention it **clearly**.

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

14

1) Match **List – I** (Studies) with **List – II** (Purpose) and select the correct answer using the codes :

List – I

- A) Economic studies
- B) Financial studies
- C) Traffic studies
- D) Engineering studies

List – II

- 1) For road location and alignment
- 2) For population and agricultural pattern
- 3) For ascertaining the source of income
- 4) For traffic volume and traffic flow patterns

Codes :

	A	B	C	D
a) 1	2	3	4	
b) 2	3	4	1	
c) 3	4	2	1	
d) 1	3	2	4	

- 2) The corrected modulus of subgrade reaction for standard diameter plate is 6kg/cm^3 . What would be the modulus of subgrade reaction of the soil when tested with a 30 cm diameter plate ?
- a) 15kg/cm^3 b) 25kg/cm^3 c) 30kg/cm^3 d) 60kg/cm^3
- 3) Speed regulations on roads is decided on the basis of
- a) 60th percentile cumulative frequency
- b) 75th percentile cumulative frequency
- c) 80th percentile cumulative frequency
- d) 85th percentile cumulative frequency
- 4) For the construction of Water Bound Macadam (WBM) roads, the correct sequence of operations after spreading coarse aggregates is
- a) Dry rolling, wet rolling, application of screening and application of filler
- b) Dry rolling, application of filler, wet rolling and application of screening
- c) Dry rolling, application of screening, wet rolling, and application of filler
- d) Dry rolling, application of screening, application of filler and wet rolling,
- 5) Which one of these methods of tunnel construction is not suitable in rocks ?
- a) Full face method b) Compressed air method
- c) Heading and benching method d) Drift method

P.T.O.



- 6) Which one of the following methods is generally considered the best for tunnel ventilation ?
 a) Driving a drift through the tunnel
 b) 'Blowin' method
 c) 'Blowout' method
 d) Combination of 'Blowin' and 'Blowout' methods
- 7) Which type of bridge is normally the longest type' ?
 a) Beam b) Arch c) Cable stayed d) Suspension
- 8) Which type of bridge foundation is suitable when rock level close to the ground or top soil strata is strong ?
 a) Deep foundations b) Shallow foundations
 c) Pile foundation d) Well foundation
- 9) Consider the following statements,
 Excessive camber is not provided on the roads because
 1. transverse tile causes discomfort
 2. of formation of cross ruts
 3. of likely toppling over of highly laden bullock carts
 4. of higher cost involved
 Which of these statements are correct ?
 a) 2, 3 and 4 b) 1, 3 and 4 c) 1, 2 and 4 d) 1, 2 and 3
- 10) Ruling gradient on highways as per IRC in plain terrain is
 a) 1 in 30 b) 1 in 60 c) 1 in 100 d) 1 in 200
- 11) If the difference in elevation between the edges of a pavement of width 9 m and its crown is 7.5 cm, what is the camber of the pavement ?
 a) 1 in 60 b) 1 in 45 c) 1 in 30 d) 1 in 15
- 12) Bitumen grade 80/100 indicates that under the standard test conditions, penetration value of would vary from
 a) 0.8 mm to 1 mm b) 8 mm to 10 mm
 c) 8 cm to 10 cm d) 0.08 mm to 0.1 mm
- 13) In cement concrete pavements, tie bars are installed in
 a) Expansion joints b) Contraction joints
 c) Warping joints d) Longitudinal joints
- 14) Match **List – I** (Item) with **List – II** (Use) and select the correct answer using the codes :
- | | |
|---------------------------------|-----------------------------|
| List – I | List – II |
| A) O and D survey | 1) Concrete pavement design |
| B) Collision diagram | 2) Compaction |
| C) OMC | 3) Accident survey |
| D) Radius of relative stiffness | 4) Traffic survey |

Codes :

	A	B	C	D
a)	3	4	1	2
b)	4	3	2	1
c)	3	4	2	1
d)	4	3	1	2



Seat No.	
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T.E. (Civil Engineering) (Part – I) (Old-CGPA) Examination, 2018
TRANSPORTATION ENGINEERING – I

Day and Date : Monday, 10-12-2018
Time : 2.30 p.m. to 5.30 p.m.

Marks : 56

Instructions : 1) Figure on **right** indicates **full** marks.
2) Assume suitable data wherever necessary and mention it **clearly**.

SECTION – I

2. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain the role of transportation in the development of country and write the different modes of transport.
- b) The area of a certain district is 80000 sq. km and number of towns 86. Calculate the length of different categories of roads to be provided in this district by the year 2001. Overall road density aimed at 82 km per 100 sq.km area.
- c) Define Stopping Sight Distance (SSD) and calculate the stopping sight distance required on sloping gradient of – 2.5% for a design speed of 100 kmph. Assume a reaction time of 2.5 sec, coefficient of friction of 0.7 and brake efficiency of the vehicle is 50%. Also calculate the intermediate sight distance.
- d) Write a short notes on objectives of
 - 1) Super elevation and curve widening
 - 2) Transition curves.

3. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) What are the objectives of O and D study ? Enlist the methods of conducting O and D study.
- b) The average normal flow of traffic on cross roads A and B during design period are 400 and 250 PCU per hour, the saturation flow values on these roads are estimated as 1250 and 1000 PCU per hour respectively. The all-red time required for pedestrian crossing is 12 sec. Design the two phase traffic signal with pedestrian crossing by Webster's method. Sketch the phase diagram.
- c) What are the various tests carried out on bitumen ? Briefly mention the principle and use of each.
- d) Write a short notes on :
 - 1) Types of parking facilities
 - 2) Steps in bituminous mix design.

SECTION – II

4. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain 'Flexible Pavement and Rigid Pavement' and bring out the points of difference.
- b) Enumerate the construction steps of Water Bound Macadam (WBM) road.
- c) Calculate the stresses at the interior, edge and corner region of cement concrete pavement using Westergaard's analysis. Use the following data : Wheel Load $P = 4100$ kg, Modulus of elasticity of cement concrete $E = 0.3$ million kg/cm² pavement thickness, $h = 15$ cm, Poisson's ratio, $\mu = 0.15$, modulus of subgrade reaction, $K = 3.0$ kg/cm³, radius of contact area, $a = 15$ cm.

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- d) Write design steps of CBR method of flexible pavement design as per IRC guidelines. Calculate the CSA for the given data.
- 1) Type of road-two lane undivided carriageway
 - 2) Design CBR value – 5.0%
 - 3) Initial traffic on completion of construction = 300 cv per day
 - 4) Average growth rate = 6.0% per year
 - 5) Design life = 10 years
 - 6) VDF value = 2.5
 - 7) Lane distribution factor = 0.75.
5. Answer **any two** questions (7 marks **each**) : **(2×7=14)**
- a) What is scour and scour depth ? A bridge is proposed to be constructed across an alluvial stream carrying a discharge of 300 m³/sec. Assuming the value of silt factor = 1.1, determine the maximum scour depth when the bridge consist of three spans of 30 m each. Assume maximum scour depth = 2 d at noses of piers.
 - b) Describe drainage in tunneling.
 - c) State methods of tunneling in soft rock. Explain with sketch any one method.
 - d) Write short notes on :
 - 1) IRC class AA loading
 - 2) Bridge bearings.
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Seat No.	
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Set	R
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T.E. (Civil Engineering) (Part – I) (Old-CGPA) Examination, 2018
TRANSPORTATION ENGINEERING – I

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Max. Marks : 70

Time : 2.30 p.m. to 5.30 p.m.

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MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

14

- 1) Bitumen grade 80/100 indicates that under the standard test conditions, penetration value of would vary from
 - a) 0.8 mm to 1 mm
 - b) 8 mm to 10 mm
 - c) 8 cm to 10 cm
 - d) 0.08 mm to 0.1 mm
- 2) In cement concrete pavements, tie bars are installed in
 - a) Expansion joints
 - b) Contraction joints
 - c) Warping joints
 - d) Longitudinal joints
- 3) Match **List – I** (Item) with **List – II** (Use) and select the correct answer using the codes :

List – I	List – II
A) O and D survey	1) Concrete pavement design
B) Collision diagram	2) Compaction
C) OMC	3) Accident survey
D) Radius of relative stiffness	4) Traffic survey

Codes :

	A	B	C	D
a)	3	4	1	2
b)	4	3	2	1
c)	3	4	2	1
d)	4	3	1	2

- 4) Match **List – I** (Studies) with **List – II** (Purpose) and select the correct answer using the codes :

List – I

List – II

- A) Economic studies 1) For road location and alignment
B) Financial studies 2) For population and agricultural pattern
C) Traffic studies 3) For ascertaining the source of income
D) Engineering studies 4) For traffic volume and traffic flow patterns

Codes :

	A	B	C	D
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b)	2	3	4	1
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P.T.O.



- 5) The corrected modulus of subgrade reaction for standard diameter plate is 6kg/cm^3 . What would be the modulus of subgrade reaction of the soil when tested with a 30 cm diameter plate ?
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- 6) Speed regulations on roads is decided on the basis of
a) 60th percentile cumulative frequency
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- 7) For the construction of Water Bound Macadam (WBM) roads, the correct sequence of operations after spreading coarse aggregates is
a) Dry rolling, wet rolling, application of screening and application of filler
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- 8) Which one of these methods of tunnel construction is not suitable in rocks ?
a) Full face method b) Compressed air method
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- 9) Which one of the following methods is generally considered the best for tunnel ventilation ?
a) Driving a drift through the tunnel
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1. transverse tire causes discomfort
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Which of these statements are correct ?
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- 13) Ruling gradient on highways as per IRC in plain terrain is
a) 1 in 30 b) 1 in 60 c) 1 in 100 d) 1 in 200
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TRANSPORTATION ENGINEERING – I

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Marks : 56

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SECTION – I

2. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain the role of transportation in the development of country and write the different modes of transport.
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SECTION – II

4. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain 'Flexible Pavement and Rigid Pavement' and bring out the points of difference.
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- d) Write design steps of CBR method of flexible pavement design as per IRC guidelines. Calculate the CSA for the given data.
- 1) Type of road-two lane undivided carriageway
 - 2) Design CBR value – 5.0%
 - 3) Initial traffic on completion of construction = 300 cv per day
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 - d) Write short notes on :
 - 1) IRC class AA loading
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Seat No.	
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Set	S
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T.E. (Civil Engineering) (Part – I) (Old-CGPA) Examination, 2018
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Max. Marks : 70

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MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

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 - a) 60th percentile cumulative frequency
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- 2) For the construction of Water Bound Macadam (WBM) roads, the correct sequence of operations after spreading coarse aggregates is
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 - c) Dry rolling, application of screening, wet rolling, and application of filler
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- 3) Which one of these methods of tunnel construction is not suitable in rocks ?
 - a) Full face method
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 - c) Heading and benching method
 - d) Drift method
- 4) Which one of the following methods is generally considered the best for tunnel ventilation ?
 - a) Driving a drift through the tunnel
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- 5) Which type of bridge is normally the longest type' ?
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 - c) Pile foundation
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P.T.O.



- 7) Consider the following statements,
Excessive camber is not provided on the roads because
1. transverse tilt causes discomfort
 2. of formation of cross ruts
 3. of likely toppling over of highly laden bullock carts
 4. of higher cost involved

Which of these statements are correct ?

- a) 2, 3 and 4 b) 1, 3 and 4 c) 1, 2 and 4 d) 1, 2 and 3
- 8) Ruling gradient on highways as per IRC in plain terrain is
- a) 1 in 30 b) 1 in 60 c) 1 in 100 d) 1 in 200
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- 10) Bitumen grade 80/100 indicates that under the standard test conditions, penetration value of would vary from
- a) 0.8 mm to 1 mm b) 8 mm to 10 mm
- c) 8 cm to 10 cm d) 0.08 mm to 0.1 mm
- 11) In cement concrete pavements, tie bars are installed in
- a) Expansion joints b) Contraction joints
- c) Warping joints d) Longitudinal joints

- 12) Match **List – I** (Item) with **List – II** (Use) and select the correct answer using the codes :

List – I

- A) O and D survey
B) Collision diagram
C) OMC
D) Radius of relative stiffness

List – II

- 1) Concrete pavement design
2) Compaction
3) Accident survey
4) Traffic survey

Codes :

	A	B	C	D
a)	3	4	1	2
b)	4	3	2	1
c)	3	4	2	1
d)	4	3	1	2

- 13) Match **List – I** (Studies) with **List – II** (Purpose) and select the correct answer using the codes :

List – I

- A) Economic studies
B) Financial studies
C) Traffic studies
D) Engineering studies

List – II

- 1) For road location and alignment
2) For population and agricultural pattern
3) For ascertaining the source of income
4) For traffic volume and traffic flow patterns

Codes :

	A	B	C	D
a)	1	2	3	4
b)	2	3	4	1
c)	3	4	2	1
d)	1	3	2	4

- 14) The corrected modulus of subgrade reaction for standard diameter plate is 6kg/cm^3 . What would be the modulus of subgrade reaction of the soil when tested with a 30 cm diameter plate ?
- a) 15kg/cm^3 b) 25kg/cm^3 c) 30kg/cm^3 d) 60kg/cm^3



Seat No.	
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T.E. (Civil Engineering) (Part – I) (Old-CGPA) Examination, 2018
TRANSPORTATION ENGINEERING – I

Day and Date : Monday, 10-12-2018
Time : 2.30 p.m. to 5.30 p.m.

Marks : 56

Instructions : 1) Figure on **right** indicates **full** marks.
2) Assume suitable data wherever necessary and mention it **clearly**.

SECTION – I

2. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain the role of transportation in the development of country and write the different modes of transport.
- b) The area of a certain district is 80000 sq. km and number of towns 86. Calculate the length of different categories of roads to be provided in this district by the year 2001. Overall road density aimed at 82 km per 100 sq.km area.
- c) Define Stopping Sight Distance (SSD) and calculate the stopping sight distance required on sloping gradient of – 2.5% for a design speed of 100 kmph. Assume a reaction time of 2.5 sec, coefficient of friction of 0.7 and brake efficiency of the vehicle is 50%. Also calculate the intermediate sight distance.
- d) Write a short notes on objectives of
 - 1) Super elevation and curve widening
 - 2) Transition curves.

3. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) What are the objectives of O and D study ? Enlist the methods of conducting O and D study.
- b) The average normal flow of traffic on cross roads A and B during design period are 400 and 250 PCU per hour, the saturation flow values on these roads are estimated as 1250 and 1000 PCU per hour respectively. The all-red time required for pedestrian crossing is 12 sec. Design the two phase traffic signal with pedestrian crossing by Webster's method. Sketch the phase diagram.
- c) What are the various tests carried out on bitumen ? Briefly mention the principle and use of each.
- d) Write a short notes on :
 - 1) Types of parking facilities
 - 2) Steps in bituminous mix design.

SECTION – II

4. Answer **any two** questions (7 marks **each**) : **(2×7=14)**

- a) Explain 'Flexible Pavement and Rigid Pavement' and bring out the points of difference.
- b) Enumerate the construction steps of Water Bound Macadam (WBM) road.
- c) Calculate the stresses at the interior, edge and corner region of cement concrete pavement using Westergaard's analysis. Use the following data : Wheel Load $P = 4100$ kg, Modulus of elasticity of cement concrete $E = 0.3$ million kg/cm² pavement thickness, $h = 15$ cm, Poisson's ratio, $\mu = 0.15$, modulus of subgrade reaction, $K = 3.0$ kg/cm³, radius of contact area, $a = 15$ cm.

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- d) Write design steps of CBR method of flexible pavement design as per IRC guidelines. Calculate the CSA for the given data.
- 1) Type of road-two lane undivided carriageway
 - 2) Design CBR value – 5.0%
 - 3) Initial traffic on completion of construction = 300 cv per day
 - 4) Average growth rate = 6.0% per year
 - 5) Design life = 10 years
 - 6) VDF value = 2.5
 - 7) Lane distribution factor = 0.75.
5. Answer **any two** questions (7 marks **each**) : **(2×7=14)**
- a) What is scour and scour depth ? A bridge is proposed to be constructed across an alluvial stream carrying a discharge of 300 m³/sec. Assuming the value of silt factor = 1.1, determine the maximum scour depth when the bridge consist of three spans of 30 m each. Assume maximum scour depth = 2 d at noses of piers.
 - b) Describe drainage in tunneling.
 - c) State methods of tunneling in soft rock. Explain with sketch any one method.
 - d) Write short notes on :
 - 1) IRC class AA loading
 - 2) Bridge bearings.
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